



WESLEYAN UNIVERSITY-PHILIPPINES

Document Title:

CAPS FORM 201 CAPSTONE TOPIC APPROVAL

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Department

GRADUATE SCHOOL

Document No:

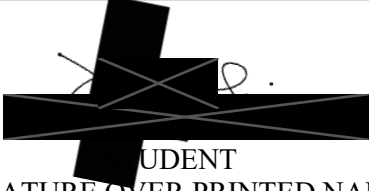
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Student's Name:		Specialization	SCIENCE EDUCATION
Degree Program:	MASTER OF EDUCATION	Date:	APRIL 11, 2026
Capstone Working Title:	Development and Validation of Digital Flip Card Material for Teaching Atomic Structure and Chemical Bonding to Grade 9 Students		
Proposed Topic:	A digital flip card material will be developed and evaluated for teaching atomic structure and chemical bonding among Grade 9 students. This study is a developmental research project that aims to design, implement, and validate an interactive flip card learning material intended to enhance students' conceptual understanding of fundamental chemistry concepts, particularly atomic structure and chemical bonding. The study is grounded on the identified need that many students struggle to understand these topics due to their abstract nature, complex representations, and lack of engaging instructional materials. In traditional classroom settings, instruction is often lecture-based and relies heavily on textbook explanations, which limits students' ability to visualize atomic models, electron arrangements, and bonding processes. As a result, learners develop misconceptions and experience low engagement when dealing with microscopic concepts that cannot be directly observed. The use of digital learning materials, such as flip cards, provides an opportunity to present information in a more interactive, visual, and student-centered manner, helping bridge the gap between abstract concepts and meaningful understanding. This project supports the United Nations Sustainable Development Goal (SDG) 4: Quality Education and SDG 9: Industry, Innovation, and Infrastructure by promoting innovative, technology-enhanced learning that improves access to effective science instruction. The integration of digital tools in teaching allows for flexible, engaging, and self-paced learning experiences that cater to diverse learners. Based on the needs analysis, students often have difficulty understanding the structure of atoms, the role of electrons in bonding, and the differences between ionic and covalent bonds. Many learners also struggle to connect symbolic representations, such as chemical formulas and Lewis structures, with actual particle interactions. To address these challenges, the project will develop a structured digital flip card material that includes key concepts, visual representations, guided explanations, and formative assessment questions designed to reinforce learning. The uniqueness of this project lies in transforming traditional flashcards into an interactive digital learning tool that promotes active recall, self-assessment, and conceptual clarity. Unlike conventional teaching strategies, this study provides a structured and research-based instructional material that enhances student engagement, improves understanding, and supports better academic performance in chemistry.		



STUDENT
(SIGNATURE OVER PRINTED NAME)

DATE

NOTED AND APPROVED BY:

PROGRAM HEAD
(SIGNATURE OVER PRINTED NAME)

DATE

GRADUATE SCHOOL DEAN
(SIGNATURE OVER PRINTED NAME)

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APPOINTED ADVISER
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